
What the Final Patch Hides: Event-Level Regression Measurement for Coding-Agent Harnesses

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Abstract

1 Coding agents are graded on the final patch alone. We instrumented an unmodified
2 coding agent to replay the benchmark’s previously-passing tests after every edit,
3 and found that transient regressions are common but almost never survive to the
4 graded patch. The instrument commits the working tree after every mutating
5 tool call and replays the benchmark’s oracle at every commit inside its pinned
6 container. We applied it to mini-swe-agent 2.4.6 unmodified, keeping its own loop,
7 prompts, model layer, budgets, and submission protocol, and replacing only its
8 environment object. On 40 SWE-bench Verified tasks it resolves 34 of 40 with
9 `claude-sonnet-4-6` and 33 of 40 with `gpt-5.6-sol`, and every run ends with a
10 changed tree. Eleven of the 80 broke previously-passing test functions, 135 counted
11 per run and 100 distinct, one still failing when the official grader read the final tree.
12 Its own testing does not close the gap. Over the window in which each test was
13 broken, no test command ran the broken test in 6 of the 10 classifiable cases. In
14 the other 4 one ran it and printed the failure by name. A method reading only the
15 agent’s logs is structurally blind to most of them. The undercount holds in our
16 harness under rollback (146 test functions broken across 8 of 47 runs, one still
17 failing) and on SWE-bench Live. Two caveats we make measurable: rollback leaves
18 clean trees partly because 17 of its 37 runs ship a tree they never changed, and a
19 model-stack difference on Verified does not survive a change of scaffold. Replacing
20 that harness’s model-written check with an execution-grounded one recovers the
21 resolve the check was destroying (13 to 26 of 40; exploratory, and circular by
22 construction). The instrument has its own blind spot. Nine of the ten trees that
23 killed the suite recorded no event. We will release the instrument, its validation
24 protocol, and the archived timelines. Appendix A catalogues 28 measurement
25 defects met while building it, 23 of which produced plausible numbers rather than
26 errors.

27 1 Introduction

28 Benchmarks for coding agents grade the final patch. SWE-bench [1] and its Verified subset [2] apply
29 the submitted patch and report whether the target tests pass and the previously passing ones still pass,
30 and work on agent-caused regressions counts what the submitted patch breaks [8]. That convention
31 cannot see what happens while the agent works, because an agent that breaks a test at step three and
32 repairs it at step five submits a clean patch. Two questions are left open: how often do agents break
33 existing behaviour during a run, and how much survives?

34 The answer was smaller than we expected, and not in the place we expected. Agents break the
35 oracle often and almost always repair it, so what the endpoint misses is transient and not shipped.

36 We measure the timeline instead of the endpoint. Every mutating tool call becomes a commit on a
 37 git reference the agent cannot see, and the oracle is re-run at every commit inside the benchmark’s
 38 pinned container. An agent operating system needs this observation primitive before it can build a
 39 recovery primitive. We contribute:

- 40 1. **An instrument for event-level regression measurement**, applicable to an agent we did
 41 not write. Every mutating tool call becomes a commit on a hidden git reference, and the
 42 benchmark’s previously-passing oracle is replayed at every recorded state inside its pinned
 43 container (Section 3).
- 44 2. **The measurement on an unmodified public scaffold**. mini-swe-agent 2.4.6 keeps its own
 45 loop, prompts, model layer, budgets, and submission protocol. Only its environment object
 46 is ours. It resolves 34 and 33 of 40 under two model families and breaks 135 oracle test
 47 functions counted per run (100 distinct) across 11 of the 80 runs, one still failing at grade
 48 time. In 6 of the 10 classifiable cases its own test commands never ran the test it broke
 49 (Section 5.1).
- 50 3. **The same measurement across recovery policies**. The undercount holds in our harness
 51 and on SWE-bench Live. But nearly half the rollback arm’s runs ship a tree they never
 52 changed, so a clean final tree partly means no edits at all (Sections 5.2, 5.3, 5.5).
- 53 4. **Recovery policy depends on the monitor’s precision**. With a model-written check, rollback
 54 buys 4 fewer contaminated trees for 11 fewer resolved instances (1 against 5; 13 against
 55 24). Replacing that check with the repository’s own oracle gets both: 26 resolved, no
 56 contaminated tree, and a lower cost per resolved instance than either. It is exploratory and
 57 circular by construction (Section 5.4).
- 58 5. **A catalogue of 28 measurement defects** found while building the instrument (Appendix
 59 A); 23 produced a plausible number rather than an error.

60 All measurements are exploratory, on development slices excluded from future confirmatory study.

61 2 Background and related work

62 **Benchmarks and their validity**. SWE-bench [1] grades a patch by tests that must go from failing
 63 to passing and tests that must keep passing. Verified [2] is the human-screened subset, deprecated
 64 as a capability metric [45]. Concerns about it have accumulated. They include contamination and
 65 memorisation [6], plausible-but-wrong patches overstating resolve by 6.2 points [36], and insufficient
 66 tests, which UTBoost [7] augmented to expose mislabelled patches affecting 24.4% of leaderboard
 67 entries. Rolling [3, 5], private [4], and long-horizon [23] alternatives came in response. TDAD [8]
 68 measured regressions in *submitted* patches on Verified (6.1% of previously-passing tests, 30.2% of
 69 non-empty patches) and reduced them with an impact-analysis map. Process-level benchmarks score
 70 intermediate states from logs [34, 35], and AgentLens [37] names regression cycles in 10.7% of
 71 passing OpenHands runs, against our 11 of the 38 runs that edited substantially (29%). A log-derived
 72 rate is bounded below by what the agent chose to run, and in 6 of our 10 classifiable cases it ran
 73 nothing that would have executed the broken test. We execute the oracle at every mutating call
 74 instead.

75 **Agent scaffolds and recovery**. SWE-agent [9], OpenHands [10], Agentless [11], and mini-swe-agent
 76 [12] span the scaffold design space [13, 14]. Within-episode recovery appears as self-reflection [15,
 77 16], progress-gated recovery [17], checkpoint repair [18], and aligned checkpoints [40]. Provenance-
 78 based recovery is surveyed in [19]. Closest to us, Kim et al. [38] match logged edits against the
 79 gold diff and recover agents that reach a gold-identical patch and then destroy it. Gao et al. [39]
 80 bind verifier evidence to code states. Neither counts regressions or compares policies. Running
 81 the repository’s tests during a run is deployed practice [42, 43]. Operating-system framings [20]
 82 and the branch-context primitive [41] supply scheduling and fork, commit, and abort, but do not
 83 observe the tree between actions. The regression problem is program repair’s plausible-versus-correct
 84 patch problem [26, 28, 29], where regression tests are the main defence against overfitting [27].
 85 Trajectory-level evaluation [21, 22] argues resolve rate alone is uninformative. Public leaderboard
 86 trajectories record actions but not the tree at each step, which the instrument commits.

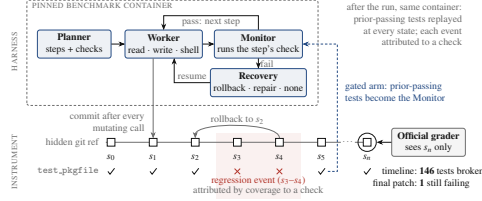


Figure 1: The harness and the instrument. Every mutating tool call commits the tree to a hidden git reference. After the run, the instance’s previously-passing tests are replayed at every state. The test row shows one regressing at s_3 – s_4 and repaired by a rollback to s_2 , which the official grader, reading only s_n , never sees. The counts are the GPT-5.6 rollback arm’s, over all 47 of its runs. In the gated arm (dashed), the timeline’s tests serve as the monitor.

3 Instrument

Figure 1 shows the harness and the instrument. **Observational timeline.** After every mutating tool call the working tree is committed to a git reference outside the agent’s view (a private index keeps its own `git status` unchanged). Rollbacks and the run’s end are observations too, and an archived run can be re-measured without re-running the agent. The agent’s tools and the harness’s checks execute inside the same pinned container as the replay, synchronised to the host tree the timeline records (Appendix A.2).

Exhaustive replay. For each observation the instance’s previously-passing (`PASS_TO_PASS`) tests run inside the pinned container against that observation’s tree. A regression event is a test that passes at one observation and fails at a later one. We replay every observation instead of bisecting, because a recovery policy makes verdicts non-monotone. Replay is scoped to the files holding those tests, which keeps it affordable on Verified (4 to 8 seconds per observation, about 26 CPU-minutes per sweep). Live’s larger oracles cost tens of minutes per instance. Infrastructure failures during replay, and tests already failing in the unmodified image, are recorded as missing observations, not failures.

Validation. Five checks run before any paid experiment (negative and positive controls, the latter including *recovered* regressions, a flake screen, an unknown-rate ceiling, a baseline liveness check), plus a golden check driving the gold patch through the agent’s real tool path and a re-scoring control that re-measures an archived timeline with no model calls. Re-scoring four arms’ bearing runs reproduced their episode counts and probe sets exactly. Appendix C gives the detail.

4 Experimental setup

Benchmark and slice. SWE-bench Verified is a measurement substrate here, not a leaderboard. Its pinned images and oracles make the measurement possible. A 40-instance development slice (16 from `django`), stratified and fixed before any measurement, was used throughout and is excluded from future confirmatory study. The GPT-5.6 stack also ran its first 10 as a calibration, and plain rollback ran twice. SWE-bench Live [3], built monthly from live repositories, is the second substrate, graded by its own harness whose rules we mirror. Its slice was fixed before any outcome (Appendix B) and its oracles are about twenty times larger (median 1,189 against 58 tests).

Harness. A planner decomposes the task into steps with verification commands, a worker executes each with three tools (`read`, `write`, `shell`), and a monitor runs the verification. On failure the recovery policy acts: **rollback** (reset to the last verified checkpoint and retry), **repair-in-place** (retry from the failed tree), or **no recovery** (the `FAIL` is final and the run ends). Runs have a \$4 work-cost cap. One instance under one policy is a cell. Each (stack, policy, substrate) triple is an arm.

Public scaffold. `mini-swe-agent 2.4.6` [12] runs unmodified (its own loop, prompts, model layer, 250-step and \$3 caps, and submission protocol) with only its environment object replaced, so every bash command executes in the benchmark’s pinned container and the tree is committed to the hidden reference after each. The seam is inert but not invisible. The agent’s `git diff` and `git status` show its own uncommitted edits and none of the instrument’s commits, but the synthetic baseline carries a marker name 54 of the 80 runs saw, and we cannot settle here whether that changed behaviour. Six departures from its leaderboard configuration, a grid census putting its sampling inside our arms’

range (a median of 3 and 2 tree-changing observations per run against our 2 to 5, at 9 to 15 bash commands each), and the rest are in Appendix C. We claim no equivalence to its published resolve rate, and because its budget differs in kind from our cap, no comparison between harnesses is tested.

Models. Two frontier stacks under identical harness, policy, instances and caps: `claude-opus-4-7/claude-sonnet-4-6` and `gpt-5.6-sol/gpt-5.6-terra` (planner/worker).

Outcomes. Resolve is the official grader’s verdict on the final patch. Events are reported at three levels because a few runs produce most: incidents (observations at which at least one oracle test broke, the co-primary endpoint), declared events (one per test function and onset, parametrised variants collapsed), and bearing runs (runs with at least one event). Incident *exposure* is a run’s incident count. An event and a final-state failure are different objects, so the undercount is in matched units (test functions broken during a run against those still failing at the end). Contamination counts oracle tests failing in the graded patch, net of baseline-dead. A tree that kills the suite is graded as failing every test but is a missing observation for event counting, so a contaminated cell need not be a bearing run.

Pre-declaration. The event unit, the proceed criteria, and the recovery-policy contrast were declared before the relevant data existed, and the analysis was committed before either comparison arm finished. One amendment preceded unblinding and one grading correction followed it (Section 5.3, Appendix B).

5 Results

5.1 An unmodified public scaffold breaks tests and puts them back

mini-swe-agent 2.4.6 ran unmodified on the same 40 instances. With `claude-sonnet-4-6`, 34 of 40 resolved (35 rot-aware) over 2,101 bash commands for \$31.38. Four runs hit its own \$3 cap. Three of those still grade resolved, because resolve reads the final tree, not the patch the scaffold never submitted. With `gpt-5.6-sol`, 33 of 40 resolved (34) over 937 commands for \$12.13. None of the 80 ended byte-identical to its start.

The timeline recorded 56 events in 5 of the 40 Claude runs and 79 in 6 of the 40 GPT-5.6 runs, on 8 instances, 3 bearing under both. Those 11 runs broke 135 oracle test functions counted per run (100 distinct, because the stacks overlap on the three shared instances). The official grader, reading only the final tree, caught exactly one of them: `test_pkgfile` in `pytest-6197` (Claude 0 of 56, GPT-5.6 1 of 79). The scaffold has no monitor or rollback, so every closure is an ordinary forward edit. The eleven stayed open for 1 to 22 bash commands (median 4), and three runs supply 81% of the 135. Against exposure rather than attempts the rate is higher. No run making two or fewer tree-changing observations was bearing, and 11 of the 38 that made more broke something (29%). Much of what the endpoint misses is a transient inside one logical edit, but transients are not free: 14% of what these runs cost (\$1.29 of \$9.52) was spent while a regression was open.

One run in full. On `django-12965` the GPT-5.6 scaffold read `SQLDeleteCompiler` for twelve commands, then at command 13 changed `single_alias` to count active aliases with `<= 1`. Its own reproduction script crashed, and at command 15 `runtests.py delete` printed `ERROR: test_fast_delete_large_batch`. Command 16 replaced the edit, command 18 ran the suite clean, and the run graded resolved at 52 of 52. The timeline holds all of this. The graded patch holds none of it.

Would its own testing have caught it? For each breaking edit we read the trajectory over the window the test was broken, from the breaking command to just before the repair, asking whether a test runner ran, whether any run would have executed *this* test, and whether its output showed the failure. Coverage is three-valued, with *unknown* for runs that stopped early or were truncated (Appendix C). In 6 of the 10 classifiable cases the agent’s own test commands never ran the broken test. In the other 4 they ran it and printed the failure by name, and it was repaired anyway. The eleventh is unknown, a `pytest -x` run that stopped earlier. Self-testing is blind to the first group and ignored on the second. The endpoint sees neither. With eleven cases this is a case series, not a rate.

5.2 The same undercount in our harness

Eight of the 47 GPT-5.6 rollback runs (the sweep and a 10-instance calibration) broke a previously-passing test, 146 test functions counting each run separately (145 distinct). One of those,

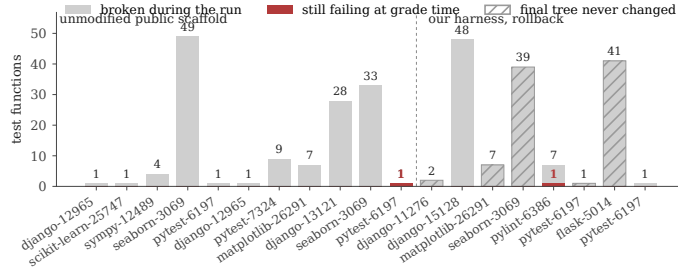


Figure 2: Every run that broke an oracle test, in matched units. Test functions broken on the timeline, dark where still failing at grade time. Left, the unmodified public scaffold under both models. Right, our rollback arm, hatched where the final tree is byte-identical to its start.

177 `test_can_read_toml_env_variable` in `pylint-6386`, was still failing at grade time. Three storms
 178 supply 88% of the 184 declared events over all 47 runs. Table 1’s 140 is the sweep alone, and declared
 179 events are exposure only. Figure 2 shows every bearing run on both sides.

180 **Attrition and rates.** Under rollback the Claude stack resolved 13 of 40 (32.5%, exact 95% CI 18.6%
 181 to 49.1%) for \$34.24 and the GPT-5.6 stack 13 of 40 for \$8.14. A circuit breaker stopped three cells
 182 and a synchronisation defect lost two, so 37 were attempted and 35 graded (13 of 35, 37.1%). Rates
 183 are out of 40 unless stated, and one instance is unresolvable under the current parser (Appendix A.3),
 184 putting the ceiling at 39 for every arm, the scaffold included. The median rollback run makes 4
 185 mutating tool calls under GPT-5.6 and 3 under Claude.

186 A clean tree does not always represent work. Excluding the hidden reference, 17 of the 37 rollback
 187 runs end byte-identical to their start, as do 19 of the Claude stack’s 40, 4 under the gate, and 0 under
 188 repair-in-place. Four of the six bearing sweep runs shipped nothing. Over the two that delivered
 189 a patch, 1 of 55 broken test functions is still failing, so the claim rests on the public-scaffold arm,
 190 where no run ends unchanged.

191 Event frequency also differed by model stack, but does not generalise. Under the same harness and
 192 policy the Claude stack produced no events in 40 runs against the GPT-5.6 stack’s 140 in 6 of 37 (one-
 193 sided Fisher $p = 0.010$; reassign three bearing runs and $p = 0.11$), yet the same `claude-sonnet-4-6`
 194 worker under unmodified mini-swe-agent is bearing on 5 of 40 with 56 events, and the difference is
 195 absent on Live. Model and adapter are confounded, and controls are in Appendix C.

196 A break that stops the suite from being collected opens no timeline event, because a test the parser
 197 never mentions counts as a missing observation, not a failure, while official grading scores such a
 198 patch as failing everything. Nine of the ten suite-killed trees across all arms recorded no event, so the
 199 two instruments miss different things.

200 5.3 Recovery policy: what the clean final tree costs

201 On the primary endpoint, contamination paired by instance, no recovery was worse than rollback
 202 on 9 instances and better on 1 (exact sign test $p = 0.021$). Repair-in-place was worse on 5, better on
 203 1 ($p = 0.22$). Incident exposure, the co-primary, did not differ ($p = 0.45$ and 1.0). The arms cannot
 204 show the policy caused the difference. No recovery ends at the monitor’s first FAIL, so its regressions
 205 arrive on the last tree-changing observation with nothing after them, and its 19-of-19 persistence is
 206 the stopping rule as much as the policy.

207 Rollback pays for its clean tree in delivered work, resolving 13 of 40 against 24 and 22, losing nine
 208 discordant pairs to repair-in-place and winning one (McNemar $p = 0.021$). The loss is the monitor’s.
 209 Of the 18 failed rollback runs, 15 failed at the first step, and 9 of those instances were resolved by an
 210 arm that kept work the check rejected and the grader accepted, and 17 of 37 runs ended unchanged.

211 **Disclosure.** The first unblinding gave $p = 0.375$, because seven cells whose final trees killed the suite
 212 had been dropped as ungradable. Correcting the grader and re-grading those archived trees gives the
 213 0.021 above, a rule change made after the data were seen (Appendix B).

Table 1: Every arm on both substrates. **contam.**: graded cells failing an oracle test, net of baseline-dead. **empty**: runs ending byte-identical to their start. **events**: declared (test function, onset) pairs, exposure only, never a denominator. **inc.**: incidents, the pre-declared co-primary. **broken** $\rightarrow$ **left**: oracle tests broken during the bearing runs against those still failing at grade time. The Claude rollback arm’s 1 is baseline rot, not agent damage (Appendix C). The public-scaffold rows ran under mini-swe-agent’s own budget, not our cap. Composition and provenance: Appendix C.

substrate	arm	resolved	contam.	empty	events	inc.	bearing	broken $\rightarrow$ left	spend
Verified	mini-swe-agent, Claude	34	0	0/40	56	5	5/40	56 $\rightarrow$ 0	\$31.38
Verified	mini-swe-agent, GPT-5.6	33	1	0/40	79	6	6/40	79 $\rightarrow$ 1	\$12.13
Verified	GPT-5.6 rollback	13	1	17/37	140	11	6/37	104 $\rightarrow$ 1	\$8.14
Verified	Claude rollback	13	1	19/40	0	0	0/40	—	\$34.24
Verified	GPT-5.6 gated	26	0	4/40	70	7	5/40	62 $\rightarrow$ 0	\$10.51
Verified	GPT-5.6 split-gated	28	0	3/40	14	4	3/40	14 $\rightarrow$ 0	\$9.40
Verified	GPT-5.6 repair-in-place	24	5	0/40	84	7	7/40	83 $\rightarrow$ 4	\$14.73
Verified	GPT-5.6 no recovery	22	9	1/40	19	3	3/40	19 $\rightarrow$ 19	\$4.40
Live	GPT-5.6 rollback	0 (2)	0	25/39	156	9	6/39	147 $\rightarrow$ 0	\$10.29
Live	Claude rollback	2 (3)	1	24/39	49	6	3/39	48 $\rightarrow$ 0	\$49.80
Live	GPT-5.6 gated	1 (4)	2	9/39	280	6	3/39	270 $\rightarrow$ 0	\$11.49
Live	GPT-5.6 no recovery	4 (6)	4	3/39	30	4	4/39	30 $\rightarrow$ 30	\$8.95

214 5.4 Replacing the check removes the tradeoff

215 Replacing the planner-written check with the repository’s own oracle takes resolve from 13 of 40
216 to **26 of 40 (65.0%)** with **no contaminated final tree**, against the ungated arms’ 24 and 22, and 4
217 empty patches against rollback’s 17. A step is rejected only if a test that passed at the start now fails.
218 This fourth arm was added after unblinding and is exploratory. The guarantee is cheap. The gate runs
219 at least 4 times in the median run at 4 to 8 seconds each, so 16 to 32 seconds of wall-clock and \$2.37
220 more model spend than rollback, and per resolved instance it costs \$0.40 against rollback’s \$0.63
221 and repair-in-place’s \$0.61. It beat rollback on 35 shared instances, winning 10 discordant pairs and
222 losing 1 ($p = 0.012$), and resolved 26 against repair-in-place’s 24, with no contaminated tree against
223 that arm’s 5.

224 The gate gives a guarantee, not a rate, and it is circular. Its tests are the benchmark’s own oracle
225 (median ratio 1.00), so a clean tree follows whenever the gate saw the final tree and the grader reads
226 what the gate read. A fifth arm weakens that, reading a deterministic half of the ids (2,278) and
227 never the other half (2,585), on which the grader also rules [44]. The split is over ids, not files, so a
228 break reaching a held-out id usually reaches a watched one, and the held-out half bounds the leak
229 instead of sampling it. A file-level or non-oracle gate would settle it. It resolved **28 of 40 (70.0%)**,
230 indistinguishable from the full gate ($p = 0.73$) and better than rollback (13 to 2, $p = 0.007$), with no
231 held-out failure apart from the parser-capped instance.

232 5.5 Replication on SWE-bench Live

233 Live keeps the undercount and loses the resolve rates, at most 6 of 40 even rot-aware, so its recovery
234 contrast is uninformative. Rollback broke 147 test functions across 6 of 39 runs, none still failing,
235 against Verified’s 104 across 6. Live’s oracles are twenty times larger, so the two counts are not
236 comparable. No recovery left all 30 of its own ($p = 0.125$). The stack difference did not replicate (3
237 of 39 against 6, $p = 0.24$).

238 6 Discussion

239 **Why timeline regressions matter.** A repaired regression still costs money. Work done while one
240 was open took 14% of the scaffold’s bearing-run cost and 27% of our harness’s (\$0.63 of \$2.33), both
241 small-base estimates. Patch verifiers [24] and process reward models [25] need step-level ground
242 truth, and instance-scale training sets [32, 33] do not carry it. The events supply it, and they tell
243 a recovery primitive [41] when to abort. Two rules would have prevented most of Appendix A’s
244 defects: record infrastructure failures as missing observations, and require positive evidence that the
245 measurement path is live. **What to build differently.** A recovery monitor should run tests, not trust a
246 model-written check. A benchmark should report what broke during the run beside the final verdict.

247 **Limitations.** Every rate estimates a 40-instance slice with one sweep per arm, and event counts
248 bound the truth from both sides (Appendix C).

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A The measurement defect catalogue

A.1 The catalogue by mechanism

Each row is a defect found while building the instrument. **S**: it produced a plausible number rather than an error. **G**: it was present while the test suite passed (A4 is the suite). **H**: it was findable only on a clean host. Class A rows depend on the machine the code runs on; class B rows render an infrastructure failure as a measurement; class C rows measure something other than the event as defined; class D rows consume a producer’s output without checking the producer.

#	Defect	Consequence	S	G	H
A1	Shadow commits inherited the machine’s git identity	timeline silently empty on any clean machine	✓	✓	✓
A2	Validation probe invoked bare python	every probe exit 127 on a clean host	✗	✓	✓
A3	Unpinned SDK resolved a different major version	two machines ran different code	✗	✓	✓
A4	Test fixtures verified with bare pytest from PATH	15 tests fail on a clean host as harness defects	✗	—	✓
A5	A benchmark scorer invoked bare python	score 0.0 from a missing interpreter	✓	✓	✓
A6	Agent executed on the host; measurement in the container	26 of 28 zero-step runs; see A.2	✓	✓	✓
B1	Output marker used shell :	a 13/13 run scored as 13 errors	✓	✓	
B2	Results on stderr, markers on stdout	one framework’s results outside the parsed slice	✓	✓	
B3	Probe diffed against a deleted commit	every graded test a hole	✓	✓	
B4	Interrupted mirror clone accepted with zero commits	later instances fail far from the cause	✓	✓	
B5	Container workdir unvalidated	every exec exit 127	✓	✓	
B6	Isolation removed loopback, not only the internet	23.5% of the oracle dead	✓	✓	
B7	Provider cache served removed containers	later cells replay against nothing and score clean	✓	✓	
B8	Tree reset reverted image build-time edits	one repository family’s oracle dead	✓	✓	
C1	Bisection assumed monotone verdicts	recovered regressions invisible	✓	✓	
C2	Declared event unit not implemented	rate off by up to 2.7×	✓	✓	
C3	Rollbacks produced no observation	recoveries invisible to the timeline	✓	✓	
C4	"Persists past the step boundary" undefined	one reading excludes the events of interest	✓	✓	
D1	Audit field never populated	0 tool errors reported, always	✓	✓	
D2	Malformed planner output crashed the run	33% of runs lost as task failures	✗	✓	
D3	Dependency install counted as agent work	attribution vacuous	✓	✓	
D4	Join took the first observation, not the last	failures pinned to the wrong tree	✓	✓	
D5	Executed config rebuilt from a name	manifest described a different run	✓	✓	
D6	Git handles never released	sweeps die after ~400 cells	✓	✓	
D7	Console re-walked the tree per run	presented as a dead server	✗	✓	
D8	Detection events lacked the ids attribution joined on	attributed detection always unknown	✓	✓	
D9	Absent coverage rendered as measured silence	unmeasured regressions reported as silent	✓	✓	
D10	Output-capped turns executed	half-written file; a 78-event storm	✓	✓	

Totals: 23 of 28 silent; 27 of 28 present under a passing suite; 6 of 28 findable only on a clean host. Ten further defects found after this census closed are described where they arose (Sections 5.2 and 5.3, Appendices A.3 and C); the rest will be released with the code.

A.2 The defect that invalidated a conclusion

After nineteen fixes, every validation check passed, the re-scoring control reproduced archived episode counts exactly, and three pilots (80 runs) measured an event rate far below the pre-declared proceed criterion. We drafted the conclusion the rule directed: the benchmark offered too little opportunity, so switch benchmarks. The conclusion was wrong. The harness ran the agent on the host in a bare source checkout, while every probe and every gate ran inside the pinned container. On the host, `import matplotlib` in that checkout succeeds by importing the uncompiled source tree as a namespace package, and everything downstream fails in ways indistinguishable from an incompetent agent. Twenty-six of 28 zero-step runs traced to this. The checks had validated the measurement path and never the agent’s execution path. The fix was to route the agent’s tools and checks through the same container as the replay, and to add a per-cell environment parity check that runs before any model call.

361 A.3 Mechanisms in public infrastructure

362 Class A: OpenHands issues #4235 and #7044 [31], environment construction failures against SWE-
363 bench images. Class B: UTBoost’s finding that the held-out oracle is insufficient at leaderboard scale
364 [7], and, on SWE-bench Live, baseline tests that fail in the unmodified image because the calendar
365 passed a deprecation date encoded in the instance. Class C: final-state regression measurement [8].
366 Class D: the SWE-bench grader accepting forged test output on stdout [30], and, at the harness’s
367 current revision, a parser that keys parametrised ids by their full text while the dataset stores them
368 truncated at the first space (the earlier parser’s behaviour), so 16 previously-passing ids of one instance
369 in our slice match no output and every submission to it grades as unresolved.

370 B Pre-declaration and analysis

371 The event unit is one (test function, onset observation) pair with parametrised variants collapsed. The
372 gates for proceeding on a substrate were an event rate of at least 0.30 per run and a bearing fraction
373 of at least 25%. Both stacks failed the conjunction (bearing 0% and 16.2%). We then re-declared the
374 gates per (substrate, model stack) before the contrast was unblinded, so no arm here is confirmatory.
375 The contrast’s primary endpoint is final-state contamination, paired by instance, exact two-sided
376 sign test. The co-primary is incident exposure. The analysis script was committed before either
377 comparison arm finished. The regression-gated arm was added after the contrast was unblinded and
378 is exploratory. No multiplicity adjustment was planned or applied. Budget exhaustion is scored as
379 failure. Instances whose baseline oracle records failures in the unmodified image are excluded from
380 resolve-rate comparability claims, and their dead tests are typed as missing observations for event
381 counting.

382 **Compute.** Every run, replay, and grade ran on one 12-vCPU, 22 GB x86 virtual machine running
383 Docker, with no GPUs. The six harness arms on Verified in Table 1 cost \$81.42 in model spend and
384 the two public-scaffold arms \$43.51, plus \$6.87 for the second rollback sweep. Calibration and pilots
385 cost a comparable amount, and the Live sweeps report their own spend in Table 1.

386 **Slice construction on SWE-bench Live.** Fixed before any outcome: pytest-parsed instances with
387 27 to 3,000 previously-passing tests, at most four per repository, earliest first, then the first 40 of 48
388 candidates whose image passed a model-free environment check. No coverage maps are built for
389 Live, so attribution there is unknown.

390 **Attribution (Verified only).** A detected regression is classified three ways: *attributed* if a failing
391 harness check and the broken test both exercise a file the agent changed at that observation (per-
392 instance coverage map at the base commit), *co-occurring* if some check failed while the regression
393 was open without such a link, and *unknown* if the map cannot say. Of the calibration’s 57 raw
394 episodes, 39 were attributed to a failing check, 16 co-occurred with one, and 2 were silent. Of the
395 sweep’s 162, 48 had none. The public-scaffold arm has no monitor, so attribution is undefined and no
396 silence statistic is reported.

397 **Grading correction between unblindings.** The first unblinding of the recovery-policy contrast gave
398 $p = 0.375$ for the primary endpoint, because seven of the eight cells whose final trees made the suite
399 uncollectable (five no-recovery, two repair-in-place) were dropped as ungradable. Official grading
400 scores such a patch as failing every test, so the most contaminated states had been dropped from the
401 endpoint they bore on most. The grader, not the replay, now distinguishes an environment failure
402 from a patch that kills the suite, and we re-graded the seven archived trees without re-running any
403 agent.

404 C The public-scaffold arm

405 **The seam.** mini-swe-agent’s loop touches the working tree in one place:
406 `DefaultAgent.execute_actions` calls `env.execute(action)`. We replace the environ-
407 ment object and nothing else, so the scaffold keeps its own loop, prompts, litellm model layer, step
408 and cost limits, and submission protocol. Our environment runs each bash command inside the
409 benchmark’s pinned container, pulls the resulting tree back to the host the timeline records, and
410 commits an observation when the tree changed. We measured two properties of the seam instead of
411 assuming them. First, the instrument does not advance the container’s git state, so the agent’s own

412 `git status`, `git log`, and `git diff` show its uncommitted work and none of the instrument’s
413 commits. The first paid pilot ran before this was true, and it spent eight of twenty commands
414 discovering that its edits had been committed underneath it. Second, the instrument is inert but
415 visible. The synthetic baseline commit it creates carries a fixed marker name, and 54 of the 80 runs
416 saw that name in some command’s output, 8 of them through `git status` (departure 2). We cannot
417 settle here whether seeing it changed behaviour. The 54 runs that saw it resolved 43 and were bearing
418 on 10, and the 26 that did not resolved 24 and were bearing on 1 (one-sided Fisher $p = 0.069$). A run
419 that edits more has both more to see and more to break, and that explains part of this, since no run
420 with two or fewer tree-changing observations was bearing at all under either condition. But within
421 the runs that edited more the association does not vanish (10 of 27 against 1 of 11, $p = 0.088$), so we
422 report it as an open question about the seam and not a settled confound. We also wrap the routed
423 command so that a command ending in a here-document terminator or a comment parses as the
424 scaffold’s own `bash -c` would parse it. Before that fix, 39 of 40 GPT-5.6 cells hit a shell syntax
425 error, because that model writes every script as a here-document.

426 **Departures from the scaffold’s leaderboard configuration.** (1) The measurement container has
427 no network, while its own deployment uses Docker’s default bridge. The unchanged prompt tells
428 the agent it may install tools, and here it cannot. (2) The container’s git history is a single synthetic
429 commit, because the image’s upstream history can contain the fix the task asks for. (3) Resolve is
430 graded on the final tree by the official grader, not on the scaffold’s submitted patch. (4) Our replay
431 and grading run outside its loop. (5) The two models are reached through the scaffold’s `litellm` layer,
432 GPT-5.6 through its `Responses-API` model class, which the `chat-completions` class cannot serve (it
433 returned three replies with no tool call and the scaffold exited on repeated format errors). (6) Prices
434 come from our verified table, because `litellm`’s carried a stale entry for one model. For these reasons
435 we claim no equivalence to its published resolve rate.

436 **Sampling grid.** Per run, median tree-changing observations and mutated files, with the run and final
437 observations excluded because neither is an opportunity to see a transition: public scaffold 3 and 3
438 (Claude), 2 and 2 (GPT-5.6); no recovery 2 and 2; repair-in-place 4 and 4; rollback 5 and 7; gate 4
439 and 4. Observations at which more than one file changed are 0 to 6% in every arm. The scaffold
440 issues 14.8 and 9.1 `bash` commands per observation against the harness’s one tool call per observation,
441 because most of its commands only read.

442 **The self-test classifier.** For each episode we take the window in which the test was broken (the
443 breaking command up to, but not including, the command that repaired it) and label three things over
444 the scaffold’s own trajectory. *Tested*: a test runner ran and its output proves it ran, so a runner that
445 died on an import does not count. *Covered*: one of those runs would have executed this test, decided
446 on the runner’s positional arguments matched at a label boundary (a run of two sibling modules does
447 not cover a test in a third), honouring `-k`, `-m`, `--deselect`, and explicit node ids. *Shown*: a covering
448 run’s output names this test on a line that also carries a failure marker. Coverage and display are
449 three-valued. A run that stopped at an earlier failure under `-x`, or whose output was truncated by a
450 pipe, is *unknown*, not negative, because calling it negative would assert that the agent ran the test and
451 was not told, from evidence that cannot support it. We audited the classifier’s labels by hand against
452 the raw trajectories of eight bearing cells from an earlier pilot of this arm, run before the two shell
453 defects below were fixed and so not among the runs reported here. The audit disagreed with an earlier
454 version of the classifier on all three totals, and the version reported here reproduces the audit exactly.

455 **The gate’s two contaminated Live trees.** One run hit the \$4 cap before the gate saw its final tree.
456 The other passed every check yet fails 13 grader tests, 7 in the file the hidden test patch rewrites.

457 **Uses.** The per-observation verdicts are the step-level ground truth that patch verifiers [24] and process
458 reward models [25] require. We release the archived timelines for that purpose.

459 **Further limitations.** Event counts miss in one direction and exaggerate in the other. A break that
460 stops the suite from being collected opens no timeline event, so the count understates what the agent
461 broke (Section 5.2). One edit can break many test functions at once, so it overstates how many
462 separate mistakes were made, which is why the undercount is reported in matched units and the
463 storms are named. Separately, no-recovery’s 19-of-19 persistence is confounded with its stopping
464 rule, since that arm ends at the monitor’s first FAIL. To separate policy from truncation we would
465 need a no-recovery arm that keeps executing after a failed check. The public-scaffold arm is one
466 scaffold at one version under two model families. Its budget differs in kind from our arms’ (\$3 billed
467 and 250 steps against a \$4 work cost), so the two harnesses are placed side by side and never tested

468 against each other. The self-test measurement rests on 11 breaking edits, one of them coded unknown.
469 Cross-stack comparisons confound model with adapter. The gated arms are post hoc and select tests
470 from benchmark metadata. Live resolve is near floor, and under rollback most Live runs ship nothing,
471 so its recovery contrast is uninformative on resolve.

472 **Bounding the model-stack difference (Section 5.2).** Exposure differs about twofold, not 140-fold.
473 In opportunity units the Claude arm makes 147 observations to the GPT arm's 218, with per-run
474 medians of 3 observations and 3 mutated files against 5 and 7. The zero is measured, not a detection
475 failure. Re-scoring that stack's earlier canary run reproduces its three events. A second plain-rollback
476 sweep reproduced the GPT side (13 of 36 graded resolved, 116 events, 6 of 37 bearing), so the
477 bearing fraction is stable even though per-instance event counts are not.

478 **The Claude rollback arm's contaminated tree.** django-14238, a run whose tree never changed.
479 Grading its unmodified base tree three times returns 38 of 39 previously-passing tests each time. One
480 graded id never appears in the log, which upstream's rule scores as a failure. It is baseline rot of the
481 kind Appendix A.3 describes, not a regression the timeline missed.

482 **The two readings.** The graded prediction excludes test files, as the official harness restores them
483 anyway, while the replay reads the tree as it stands, so an agent edit to a test file is visible to the
484 timeline and not to the grader.

485 **Reading Table 1.** *Baseline-dead* tests already fail in the unmodified image. *Rot-aware* resolve
486 counts a run resolved when every target test passes and no failure lies outside that set. On Live,
487 **resolved** is of 40 while **empty** and **bearing** are of the 39 attempted (torchtune-1806 exceeds the
488 replay budget). **broken** → **left** omits any cell whose suite the grader could not collect, since such a
489 cell would otherwise read as perfect capture, so repair-in-place's row covers six of its seven bearing
490 runs and 83 of its 84 events.

491 **Incidents.** The pre-declared co-primary, per arm, in the order of Table 1: 5, 6, 11, 0, 7, 4, 7, 3, 9, 6, 6,
492 4.

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Justification: No models are trained. Section 4 gives the slices and how they were fixed, the harness and its three recovery policies (the two regression-gated variants are specified in Section 5.4), the public scaffold and its unmodified budgets, the exact model strings, the per-run cost cap, the outcome definitions, and the pre-declared analysis.

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Answer: [Yes]

Justification: The Claude rollback arm’s resolve carries an exact (Clopper–Pearson) 95% interval; paired comparisons use exact sign tests and exact McNemar tests, and the cross-stack comparison a one-sided Fisher exact test. The tests reported and the absence of multiplicity adjustment are declared (Section 5.4, Appendix B). Plain rollback was swept twice on Verified (Section 4) but the replicate’s numbers are not reported here, and intervals treat instances as independent although 16 of the 40 Verified instances come from django.

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Answer: [Yes]

Justification: Section 3 gives replay cost per observation and per sweep; Table 1 gives model spend per arm and substrate; Appendix B states the machine (one 12-vCPU, 22 GB x86 virtual machine running Docker, no GPUs) and the total model spend including pilots.

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Answer: [Yes]

Justification: The contribution is an evaluation instrument and measurements of it, and the plausible direct impact is positive: agents whose regressions are visible break less existing behaviour, and the archived timelines give patch verifiers and process reward models step-level supervision they currently have to guess at. Two negative paths are worth naming. An instrument that commits the working tree after every action is agent telemetry: deployed outside a benchmark it would record whatever an agent touches, including proprietary or personal code, so the hidden reference is a surveillance surface as much as a measurement one. And a gate that rejects any step failing a prior-passing test is an enforcement mechanism, not only an observer: applied to real repositories it would suppress legitimate refactors that temporarily break tests, which is the behaviour a maintainer sometimes wants. Both argue for the gate being advisory and the timeline being local and short-lived by default.

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